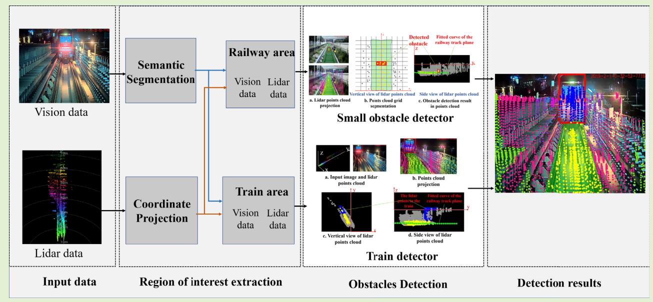
A Camera and LiDAR Data Fusion Method for Railway Object Detection

Wang Zhangyu, Yu Guizhen, *Member, IEEE*, Wu Xinkai, Li Haoran, *Member, IEEE*, and Li Da



***Abstract*—Object detection on railway tracks, which is crucial for train operational safety, face numerous challenges such as multiple types of objects and the complexity of train running environment. In this study, a multi-sensor framework is proposed to fuse camera and LiDAR data for the detection of objects on railway track including small obstacles and forward trains. The framework involves a two-stage process: region of interest extraction and object detection. In the first stage, a multi-scale prediction network is designed to achieve pixel level segmentation of the railway track and forward train via the image. In the second stage, LiDAR data is used to estimate the distance to the train and detect small obstacles in the railway track area which is extracted from the first stage. Experimental results show that the region of interest extraction method achieves desirable accuracy for railway track and train segmentation; and the proposed fusion method outperforms the one based on camera or LiDAR alone for small obstacles and forward train detection. Moreover, in practice the proposed framework has been successfully applied on the Hong Kong Metro TSUEN WAN line and the Beijing Metro YANFANG line.**

***Index Terms*— Data fusion, railway, vision, LiDAR, object detection.**

I. INTRODUCTION

SAFETY is crucial for rail transportation [1]–[3]. Most of the traffic accidents that occurred in rail transit are caused by collisions with obstacles [4], [5]. However, there is limited automatic railway obstacle detection readily applicable in the train system. Currently, whether or not there are obstacles on the railway track in front of the train can only be manually observed by the driver.

A. Background

To reduce railway accidents, numerous researches have been focused on automatic railway object detection [6]–[8], and most of the existing methods rely on vision. However, due to the complexity of the railway environment, vision-based methods face a series of challenges: (1).The first is the diversity of objects, which includes not only large-sized obstacles but small ones. Numerous collisions occur between trains or between the train and small obstacles [9]. (2).The second is the complicated running environment. Trains run on straight railway tracks, switched tracks and curved tracks. (3).Third, vision-based obstacle detection usually fails to measure the distance to obstacles accurately, especially over a long distance. Those challenges make railway object detection more difficult.

Sensor techniques, especially the LiDAR sensor, have developed rapidly in recent years [10], [11]. Since camera sensors and LiDAR sensors complement each other in terms of color and depth perception [12], it is possible to improve railway object detection by fusing camera and LiDAR data. With the high resolution and the context information, the image provides the details of the objects. Furthermore, LiDAR data provides precise depth information [13]. Therefore, the fusion of camera and LiDAR sensors is expected to achieve more accurate object detection for railways.

Accordingly, this paper proposes a railway track object detection framework which fuses the camera with the LiDAR data. To the best of our knowledge, such a framework has not been developed previously for object detection on railways. Figure. 1 presents the basic flowchart of the proposed data fusion framework. First, the image and LiDAR synchronization data which are captured by an onboard camera and a LiDAR sensor, respectively, are acquired. Then,

Manuscript received February 3, 2021; revised March 14, 2021; accepted March 14, 2021. Date of publication March 17, 2021; date of current version June 14, 2021. This work was supported in part by the Beijing Natural Science Foundation under Grant #L191002 and in part by the National Natural Science Foundation of China under Grant #52072020. The associate editor coordinating the review of this article and approving it for publication was Prof. Meribout Mahmoud. *(Corresponding author: Wu Xinkai.)*

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Digital Object Identifier 10.1109/JSEN.2021.3066714

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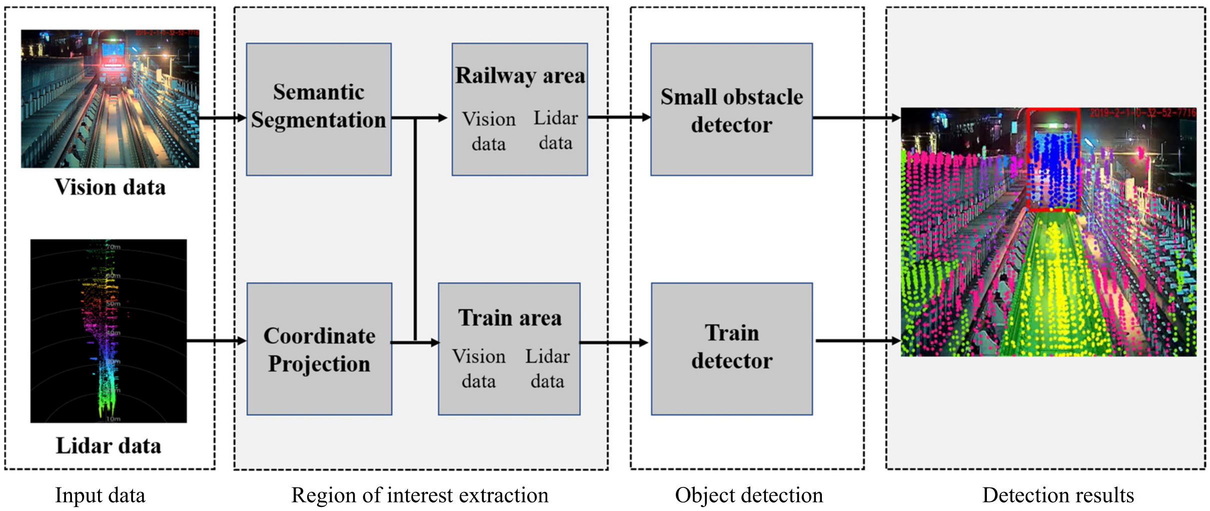

Fig. 1. The framework of the proposed method. In detection results, blue points represent the LiDAR points mapped in the forward train; yellow points represent the LiDAR points mapped in the railway track area.

the architecture is divided into two stages: the region of interest extraction and object detection.

In the region of interest extraction stage, a multi-scale prediction (MSP) network is designed to achieve pixel level segmentation of the railway track area and the forward train area. Particularly, accurate railway track area extraction can be used for the small obstacle detection, and the train area segmentation can be used for the forward train detection. Also, during the first stage we further map the LiDAR points onto the image through coordinate projection, in order to obtain the LiDAR points in the railway track area and the LiDAR points in the forward train area.

With precise classification, LiDAR points in the railway track area and in the forward train area are obtained. Then, in the second stage, two detectors are designed to detect objects, including small obstacle detector and forward train detector. For the small obstacle detector, small obstacles detection is conducted using the LiDAR points in the railway track area. We fit a railway track plane equation using the LiDAR points in the railway track area, and then search for LiDAR points higher than the railway track plane. For the forward train detector, the distance to the forward train is obtained using the LiDAR points in the train area. As the LiDAR points of the forward train area have been segmented through the region of interest extraction stage, the accurate distance to the forward train can be obtained through a clustering algorithm.

B. Contributions

The main contributions of this paper include:

(1) An efficient framework to fuse camera and LiDAR data is designed. The framework includes the region of interest extraction and object detection, and the framework can detect both large objects such as forward trains and small objects on the railway track such as stones and pedestrians.

(2) A semantic segmentation network is designed to fuse multi-scale features of the image. The network combines the encode-decode structure and the two-branch structure, and enables to achieve precise pixel level segmentation of the image.

(3) A small obstacle detection method is proposed, which first conducts railway track plane fitting, and then detects small obstacles above the railway track plane.

C. Paper Organization

The remainder of this paper is structured as follows. Section II gives a brief overview of related approaches that deal with railway objects detection, followed by the proposed region of interest extraction method in Section III. Section IV elaborates the proposed object detection method. Section V shows the experimental results. Finally, conclusion is presented in Section VI.

II. Related Work

Railway object detection has been an important research topic [14]. Generally, the methods fall into three categories: vision-based approach, radar-based approach and fusion-based approach.

A. Vision-Based Approach

The vision-based approach typically uses a machine vision method to analyze the image captured by an on-board camera, and therefore to obtain object information [15]–[17]. For example, Mukojima *et al.* proposed a background subtraction based method which compared the input image with the reference train frontal view images to identify the kinds of obstacles [5]. In the literature [7], three images feature extraction and feature analysis methods were proposed to detect objects in simple scenes. Fioretti *et al.* proposed an approach to reconstruct the trajectory of a railway vehicle and promising results were obtained for obstacle detection [16]. On the other hand, the semantic segmentation that can obtain the category information of each pixel, has been developed rapidly in recent years. Currently, the state-of-the-art semantic segmentation networks mainly include two kinds: two-branch structure and

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encoder-decoder structure [18], [19]. The encoder-decoder structure typically has a deeper network to extract features, whereas the two-branch structure is usually more condensed. Vijay *et al.* proposed the SegNet for semantic segmentation and this was the first time using the encoder-decoder structure for semantic segmentation [20]. Adam *et al.* proposed the ENet for semantic segmentation and it could be applied to embedded platforms [21]. Eduardo *et al.* proposed the ERFNet, and achieved a balance between computational efficiency and accuracy [22]. However, it should be noted that those approaches appear incapable of providing accurate estimation of distance to objects.

B. Radar-Based Approach

Unlike cameras, radar sensors that can offer accurate distance information for the detected objects, have been widely used for various applications. Radar-based approaches use radar or LiDAR to explore objects. Lüy *et al.* proposed a collision avoidance system to detect obstacles in light rail systems using a multilayer laser scanner [23]. Zheng *et al.* used a high-resolution radar and designed a double threshold generalized likelihood ratio test for robust object recognition [24]. However, it is worth mentioning that those radar-based methods may fail to identify the types of objects.

C. Fusion-Based Approach

Due to the disadvantages of single sensor, multi-sensor fusion is increasingly becoming a trend, and has inspired numerous research [25]–[27]. For example, Karaduman *et al.* used a camera and laser meter to identify objects simultaneously [9]. García *et al.* proposed a multisensory system and used infrared and ultrasonic sensors for railway objects detection [2]. However, there remains a research gap to fuse camera and LiDAR data for railway objects detection. On the other hand, fusion methods have been applied to autonomous vehicles [28]–[30]. Wahde *et al.* trained several fully convolutional neural networks to carry out road detection using LiDAR-camera fusion data as input [31]. Jianhui *et al.* acquired six-channel data through vision and LiDAR data fusion and then classify road objects [11]. Jun *et al.* proposed a fusion framework, which contained a LiDAR-based weak classifier and a vision-based strong classifier for pedestrian detection [32]. In the literature [33], a deep-learning-based approach for autonomous vehicle object detection which fuses the camera and LiDAR data was proposed and the experimental results showed significant improvements. Inspired by these methods, this study aims to apply vision and LiDAR fusion to railway objects detection.

III. Region of Interest Extraction

In our data fusion framework, region of interest extraction is used to estimate the possible location of objects. In this study, a multi-scale prediction (MSP) network is proposed to segment the railway track area and the forward train area. Furthermore, the segmentation results are refined to improve the accuracy of the output. The flowchart of region of interest extraction is shown in Figure. 2.

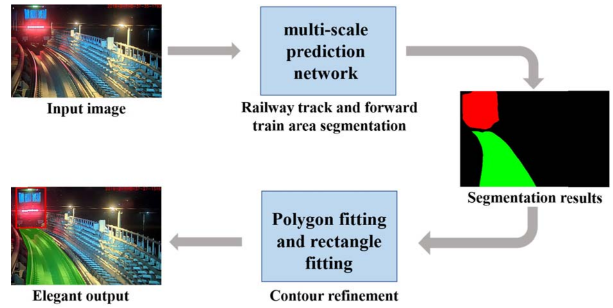

Fig. 2. The flowchart of region of interest extraction.

A. Railway Track and Forward Train Area Segmentation

In our previous work, we designed a semantic segmentation network to achieve accurate rail area detection [8], and the network can detect the railway track in various scenarios. Based on it, we further extend the network and propose an MSP network for railway track and forward train segmentation simultaneously, which help get the detection region for object detection.

Generally, the state-of-the-art semantic segmentation networks can be divided into two-branch structure and encoder-decoder structure. Encoder-decoder structure uses a series of convolution layers to extract features of the image in the encoder part and up-sampling layers to recover the spatial details in the decoder part. By comparison, two-branch structure mainly contains two embranchments with different resolutions to extract different levels of feature information. Feature map resolution and network depth are the key factors affecting running time [19]. Therefore, two-branch structures appear more appropriate for meeting real-time application requirements. However, there exists one drawback that the networks in two-branch structure are often independent of each other. That is, they cannot share shallow feature maps and make full use of shallow feature information. Two kinds of structures are illustrated in Figure. 3.

Inspired by the desirable features of these structures and the work [19] that combined an encode-decode structure and a two-branch structure, we proposed a multi-scale prediction network for railway track and forward train segmentation.

The MSP network structure is illustrated in Figure. 4. Cascade sampling layers is used to generate different scales of feature maps and predict the output on each scale directly. The feature map in high resolution contains the local characteristics of the railway track and train, and the feature map in low resolution contains the global information in the interest area. Their combination can extract both global and local features of the image. Meanwhile, in MSP structure high resolution feature maps are reused to low resolution one in order to reduce computational effort. Such enhancement differs from the two-branch structure, which predicts the outputs of different scales separately.

The feature scales used in MSP structure include 1/2, 1/4 and 1/8. Additionally, in 1/8 scale feature maps dilated convolution is used to get a larger receptive field, and thus get better global information of the railway area and the train. Dilated convolutions [22], [34] can gather more context with

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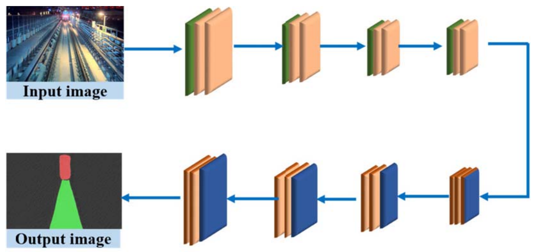
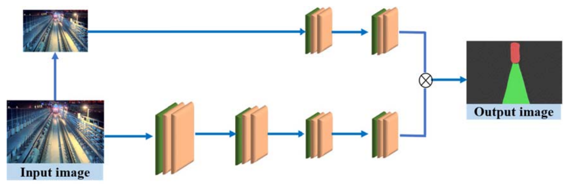


a. Encoder-decoder structure



b. Two-branche structure

Fig. 3. General semantic segmentation network structure. (a). Encoder-decoder structure; (b). Two-branch structure.

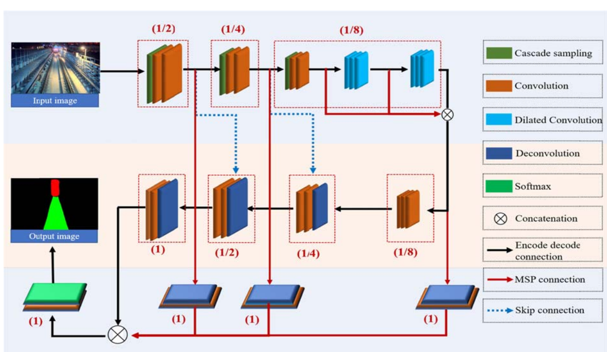


Fig. 4. Network architecture.

the resolution of feature map unchanged. In our previous work [8], the dilated-cascade connection is used to maintain the balance between global information extraction and high resolution. Thus, in multi-scale prediction structure as illustrated in Figure. 4, different dilated convolutions with rates 1, 2 and 4 are fused to explore the global information of the railway track and train.

In the last part of the network, a softmax layer is used to classify railway track and train pixels in the image. Additionally, non-bottleneck residual modules [35] are applied to help obtain higher detection accuracy.

B. Contour Refinement

Using the MSP network enables to achieve pixel-level classification of the railway track and the train in the input image. However, the outline of the railway and the train may not be smooth as illustrated in Figure. 5. Thus, it needs to refine the boundary of the railway track and the train.

An efficient polygonal fitting method [8] is applied here to fit the railway track. By using a track switch image as an example, the segmentation results of railway tracks and trains are shown in Figure. 6a. The details of algorithm 1 is elaborated as follows:

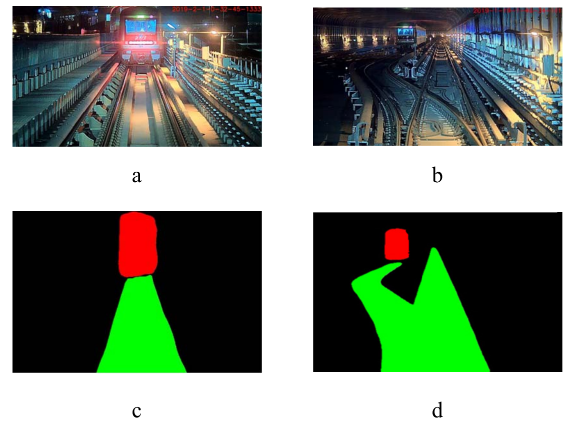


Fig. 5. Output of multi-scale prediction network. (a)-(b). The original images; (c)-(d). the segmentation results.

Algorithm 1 Polygonal Fitting

Input: The contour of the railway track
Output: Refined railway track
A: bottom leftmost point of the railway track contour
B: bottom rightmost point of the railway track contour
C: top points of the railway track contour
H: pre-set distance threshold
j: Number of boundary points

$j = 0$
for $\widehat{BI}, I\epsilon(A, C)$ **do**
  (1) From curve $\widehat{BI}$ find the farthest point from line BI, represented by point D_j.
  (2) Calculate the pixel distance between point D_j and line BI, and the distance is expressed as h.
  while $h > H$
    (3) Set D_j as the new boundary point
    (4) $j++$
    (5) Create curve $B\hat{D}_j$ and $I\hat{D}_j$, for each curve, return to (1)
end
return A,B,C, D_j, $j\epsilon(1, 2, \cdots)$

(1) Use a chain-code tracking algorithm to get the contours of the railway track as shown in Figure. 6b. Three boundary points, i.e., the bottom leftmost point, bottom rightmost point and the top point of the railway track contours, are expressed by A, B and C, respectively.

(2) From the curve $\widehat{AB}$, find the point farthest from line AB, represented as point D. Calculate the pixel distance between point D and line AB, and the distance is expressed as h. If h is bigger than the pre-set distance threshold H, set D as the new boundary point.

(3) Repeat Step 2 and continue searching boundary points between the curve $\widehat{AD}$ and $\widehat{DB}$ until all the boundary points of the curve $\widehat{AB}$ are found.

(4) Search boundary points between curve $\widehat{BC}$. Repeat Steps 2 & 3. The refined results are shown in Figure. 6c, as illustrated in red lines.

Furthermore, since the front-on view of a train is usually rectangular, rectangular fitting of the train pixels is performed.

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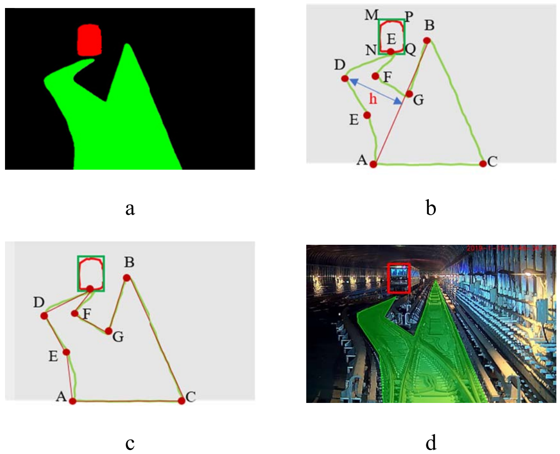

Fig. 6. Contour refinement. (a). The original segmentation result; (b)-(c). The diagrams of contour optimization process; (d). The refinement result.

As illustrated in Figure. 6b, the maximum and minimum values of the horizontal and vertical coordinates in all train pixels are estimated, and the four vertices of the train are calculated, expressed by points M, N, P, and Q. By resorting to the above method, the contours of the railway and train are refined. The refined results are shown in Figure. 6d.

Besides, as the internal and external parameters between the camera and the LiDAR have calibrated. The LiDAR points are mapped onto the image through coordinate projection [36]. Based on the LiDAR points in the railway track and the train area, object detection will be conducted in the next stage.

IV. Object Detection

By applying the MSP network, the accurate detection region in the image is obtained. Meanwhile, the LiDAR points are classified and the detection region for objects detection in LiDAR data is obtained through the coordinate unification of the LiDAR and the image. The LiDAR points in the railway track area are used for small obstacle detection, and the LiDAR points in the forward train area are used to exact the distance to the forward train. As a result, forward object detection is more precise.

A. Small Obstacles Detection

Small obstacles refer to the small objects in the railway track area such as stones or pedestrians on the rail track. These obstacles are difficult to be trained and learned by machine learning algorithms due to the limited number of samples. In subsequent experiments, three different types of small obstacles are examined, including toolkits, suitcases and pedestrians. The widths and heights of each type of obstacles are 39cm · 58cm, 42cm · 68cm, 40cm · 178cm, respectively. In this study, the LiDAR points in the railway track region are acquired via region of interest extraction. Then, they are processed to detect small obstacles in the railway track area. Note the method cannot classify small obstacles.

In order to detect the obstacles within the railway track area, it needs to fit a railway track plane equation using the LiDAR points first. As illustrated in Figure. 7, the railway track plane fitting method is described as follows:

(1) As shown in Figure. 7a, the LiDAR points are projected into the image and classified in the railway track area, as shown in yellow in raw LiDAR point cloud in Figure. 7b. (2) A grid method (to be described later to segment the point cloud) is applied to get the height information at each grid point of the railway.

(3) The Random Sample Consensus (RANSAC) algorithm [37] is applied to fit the railway track plane. The point cloud acquired from the segmentation is used to obtain the parameters of the quadratic curve for the railway plane. Thus, the height of the railway at each position can be obtained. The fitted curve of the railway track plane is shown as in green in Figure. 7d.

Figure. 7c presents the schematic diagram of the vertical view of point cloud, in which yellow points and gray points represent the LiDAR points in and outside the railway track area, respectively. It can be found that by the reflection of rails or sleepers, the LiDAR points in the railway track area are unevenly distributed. In addition, the heights of LiDAR points in the railway track area vary significantly. In such case, to fit the railway track plane equation with LiDAR points cannot guarantee desirable results. Thus, a grid method is adopted to segment and merge the LiDAR points for fitting purpose.

As illustrated in Figure. 7c, the x-coordinate and y-coordinate represent the horizontal and vertical distance of the LiDAR points, respectively. The grids are divided into the size of X · Y (0.4m·1m in this case). In row r, the height of the lowest point h_r can be calculated as follows:

$$h_r = z(\{min_z(P^i_{xyz}), \quad i \in r\}) \tag{1}$$

where $z()$ represents the height of LiDAR point, and P^i_{xyz} represents the LiDAR point in row r. The railway track plane equation is fitted according to the lowest points of each row of the grids. The curve equation of the railway track plane in RANSAC algorithm can be described as:

$$z = a \cdot y^2 + b \cdot y + c \tag{2}$$

where y represents the longitudinal coordinate information of LiDAR points; z represents the height of the railway track plane at the distance y; and a, b and c are the parameters of the fitted curve.

The fitted curve of the railway track plane is shown as the green line in Figure. 7d. Based on it, the difference between the height of each LiDAR point in each grid and the height of the fitted railway track plane can be calculated:

$$\Delta z_i = z_i - (a \cdot y_i^2 + b \cdot y_i + c) \tag{3}$$

where z_i represents the height of point p^i_{xyz}; and y_i represents the distance of point p^i_{xyz}. If Δz_i is greater than the threshold T1, this point is regarded as the outlier. Note that for most trains the height from the bottom of the train to the sleepers is 440mm.Thus, the threshold T1 is set as 440mm. Besides, in practical applications noise usually exists in the point cloud. In order to make obstacle detection more reliable, if the count of outlier points in each grid exceeds the threshold T2, this

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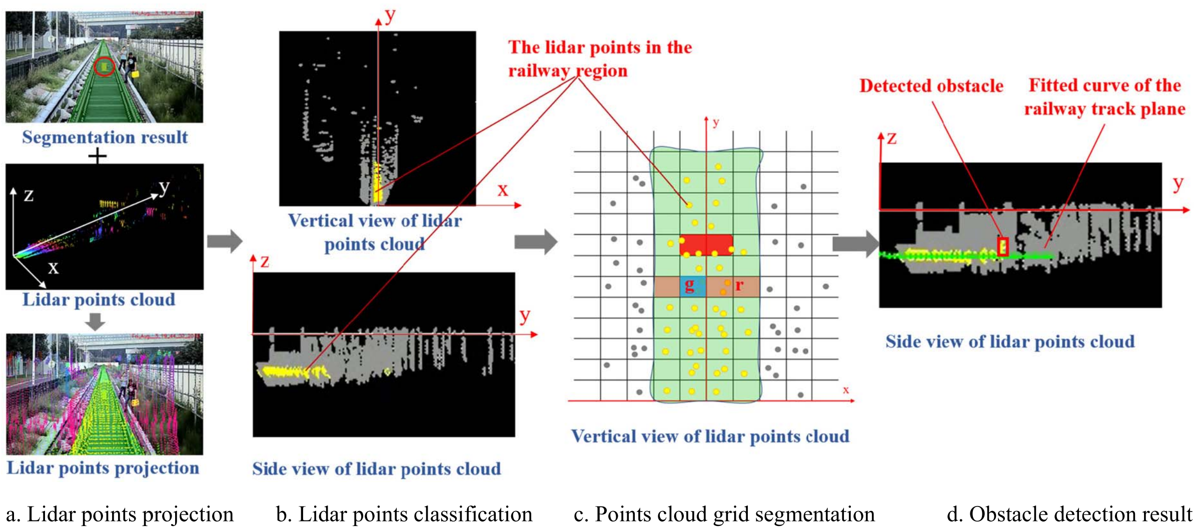

Fig. 7. Railway track plane fitting. (b). The vertical and side views of the LiDAR point clouds, in which, the yellow points are the LiDAR points that are in the railway region, and the gray points are the LiDAR points outside the railway area; (c). The schematic diagram of the vertical view of point cloud.

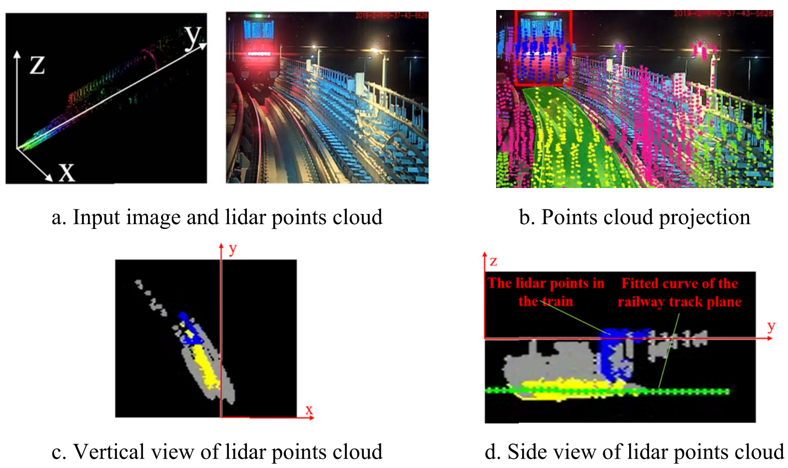

Fig. 8. Forward train detection. (a). The image and LiDAR point cloud synchronization data; (b). The fused data in which the LiDAR points are projected to the image; (c)-(d). The vertical and side views of the LiDAR point cloud; The blue points in the (c) and (d) are the LiDAR points in the train.

grid is considered as a grid of obstacles (here the threshold T2 is empirically set as 3). As shown in Figure. 7a, there is a toolbox in the center of the railway track and the detection results are shown in Figure. 7c and Figure. 7d.

B. Forward Train Detection

Based on the MSP network, the pixels of the forward train area are acquired. Then, the LiDAR information is fused to obtain the distance to the forward train.

Figure. 8a shows an image with the LiDAR point cloud synchronization data. Figure. 8b illustrates the fused data after projecting the LiDAR points to the image. Accordingly, the LiDAR points belonging to the forward train area and the railway track area can be readily identified. Figures 8c and 8d show the vertical and side views of the LiDAR point cloud. Yellow and blue points denote the LiDAR points in the railway track area and forward train area, respectively, whereas gray points denote the other ones outside these areas. Based on the LiDAR points acquired in the train area, the distance to the train can be measured. However, there exists the possibility that due to mapping errors some LiDAR points outside the train area may be mistakenly accounted for. It indicates a need in practice to refine the LiDAR points for the train.

It is supposed that the leading edge of the train can be approximately taken as a flat surface, and the LiDAR points with higher densities in the leading edge should be the ones in the train area.. Thus, the distance to the train is calculated as the average distance of the densely distributed LiDAR points in the train area. In this study, an adaptive point clustering algorithm is proposed to cluster the LiDAR points and estimate the distance to the train, as shown in Algorithm 2:

Step 1: Sort LiDAR points which are projected to the train according to the LiDAR distance information.

Step 2: Set a point cluster and calculate the distance between all LiDAR points in the train to the nearest one. The points with the distance less than the pre-set threshold will be put

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Algorithm 2 Adaptive Points Clustering

Input: Train LiDAR points which are sorted by the point distance
Output: Detected distance to the train
$T1$: threshold 1 (0.3m in this study)
n: the number of all LiDAR points
m: the number of the LiDAR clusters
$c_{\mathbf{m}}$: the count of LiDAR points in cluster m
c_i : the count of LiDAR points in cluster i
$d_{\mathbf{m}}$: the distance of cluster m
$d_{\mathbf{i+1}}$: the distance of cluster i + 1
$p_{\mathbf{i}}$: the LiDAR point i
P_i^y : the distance of LiDAR point P_i
d: the distance to the detected train
$m = 0$
for $i = 1$ to n **do**
  if $P_i^y != 0$ **then**
    $c_{\mathbf{m}} = 0$;
    for $j = i + 1$ to n **do**
      if $P_j^y != 0$ **then**
        if $|P_j^y - P_i^y| < T1$ **then**
          $d_{\mathbf{m}} = \frac{P_j^y + P_i^y}{2}$;
          $c_{\mathbf{m}} + +$;
          $P_j^y = 0$
        end
      end
    end
    $m + +$
  end
end
$d = d_0$
for $i = 0$ to $m - 1$ **do**
  if $c_{\mathrm{i+1}} > c_i$ **then**
    $d = d_{\mathrm{i+1}}$;
  end
end
return$d$

into one cluster and the count for the points in this cluster updated. Meanwhile, the clustered points will not be used in the following steps.

Step 3: For the points with the distance larger than the preset threshold, set a new point cluster and repeat Step 2 until all the points are classified into a certain cluster.

Step 4: Select the point cluster with the maximum number of points as the one for the train, and calculate the average distance of the points in this cluster as the distance to the train.

V. Experiments

A. Dataset and Setup

In this study, the Cepton HR80T as a long distance LiDAR [38] with the scanning frequency of 10Hz and an industrial camera with a resolution of $1280 \cdot 720$ were used for experiments of obstacle detection. These two sensors were installed in the locomotive of the train, and firmly set to guarantee steady perspective and performance during train operation, as shown in Figure. 9.

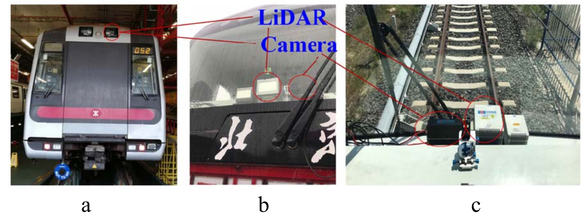

Fig. 9. Experimental platform. (a). Hong Kong Tsuen Wan line train; (b). Beijing Yanfang line train; (c). Sensor installation position.

A large amount of LiDAR and image synchronization data were collected from the Beijing Metro Yanfang line and the Hong Kong Metro Tsuen Wan line. To examine the validity of the proposed region of interest extraction algorithm, an image dataset called "Rail-dataset" was established, which accounted for typical scenarios of train operation. The Rail-dataset was labeled in Cityscapes [39] standard format and contains 10,200 finely annotated samples (7800 for training and 2400 for testing). The pixels in the dataset were mainly labeled as railway track area, train area and background.

All the region of interest extraction experiments were conducted using the open-source architecture Caffe [40] with CUDA and CuDNN backends. The loss function is the cross entropy of all the pixels in a mini-batch. The batch size of the training was set as 10. We started with a learning rate of 0.01 and a momentum of 0.9 and dropped the training rate to 90 percent every 2,000 times to accelerate convergence. Stochastic gradient descent [41] was used for parameter optimization.

B. Region of Interest Extraction Performance

For region of interest extraction, the size of the input image was $640 \cdot 360$. Two popular indicators, i.e., the Mean Intersection-over-Union (MIoU) metric and the Mean pixel accuracy (MPA) metric, were used to evaluate the accuracy of the algorithm, as shown below:

$$MIOU = \frac{1}{n+1} \sum_{i=0}^{n} \frac{TP_i}{TP_i + FP_i + FN_i} \tag{4}$$

$$MPA = \frac{1}{n+1} \sum_{i=0}^{n} \frac{TP_i}{TP_i + FN_i} \tag{5}$$

where $\mathrm{n}+1$ represents the number of segmentation categories ($\mathrm{n} + 1 = 3$ in this study); TP_i refers to true positive pixels about category i, and FN_i and FP_i refer to false negative pixels and false positive pixels about category i, respectively.

The experimental results are shown in Table I. It can be found that the MSP net achieved railway track and train region segmentation with desirable accuracy, i.e., 95.91% and 97.82% in terms of MIoU and MPA. To further investigate the effect of MSP and dilated convolution on the network, these two parts were removed in experiments for comparison purpose. The results are also shown in Table I.

It is clear that the MSP net has the highest accuracy. Without MSP, MIoU and MPA decrease by 11.04% and

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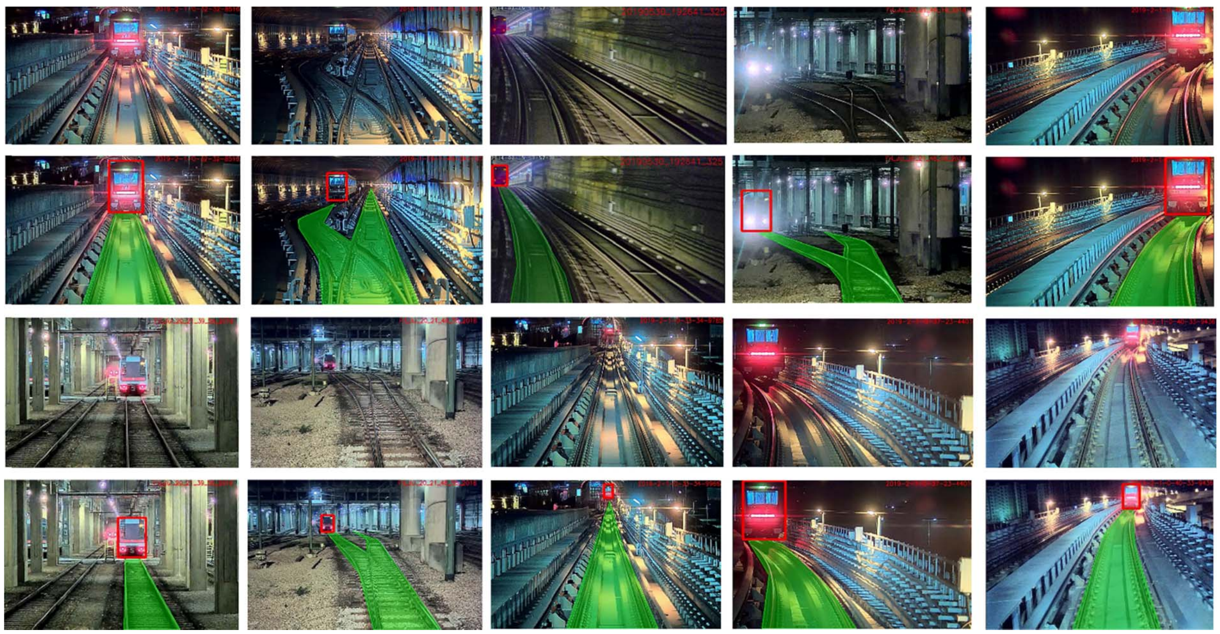
TABLE I
EXPERIMENTAL RESULTS

Method	Evaluation Metric	
	MIoU	MPA
Without multi-scale prediction	84.87%	91.64%
Without dilated convolution	90.70%	95.2%
MSP net (Proposed method)	**95.91%**	**97.82%**



a. Straight b. Switch c. Slop d. Left-turn e. Right-turn

Fig. 10. Visualization of experimental results. Row 1 and row 3 are the original images; and row 2 and row 4 are the detection results; The red bounding box represents the detected forward train, and the green area represents the detected railway track area.

TABLE II
COMPARISON RESULTS TO OTHER METHODS

Method	Accuracy Evaluation		Speed Evaluation (fps)	
	MIoU	MPA	Titan X	TX2
Segnet	84.17%	91.43%	10	1
Enet	76.82%	87.09%	66	6.6
Erfnet	91.89%	95.79%	103	10
MSP net (Proposed method)	**95.91%**	**97.82%**	**58.8**	**5.7**

6.18%, respectively. It helps demonstrate the advantage of MSP to extract multi-scale features effectively. Furthermore, without dilated convolution MIoU and MPA decrease by 5.21% and 2.62%, respectively. It reveals that the dilated convolution helps expand the receptive field. Some visualization results of railway track and train segmentation are shown in Figure. 10. It demonstrates that the proposed method enables to extract the detection region in a variety of scenarios, such as straight-track, switched-track, left-turn, right-turn track environments, etc.

Towards a more comprehensive analysis, we compared our semantic segmentation network with the state-of-the-art algorithms such as Segnet [20], Enet [21] and Erfnet [22]. In addition, in view of the requirement of real-time performance for practical application of the system, the computational efficiency was compared based on desktop GPU Nvidia Titan X and embedded GPU Nvidia Jetson tx2. The comparison results are shown in Table II.

Evidently, the MSP net outperforms the state-of-the-art segmentation networks in detection accuracy. It is essentially because our network fully considered the operational characteristics of the railway. Besides, our network achieved 58fps on Nvidia Titan X GPU. Though not the fastest (compared to Erfnet), it achieved a good balance between computational efficiency and accuracy. As the maximum scanning rate of the LiDAR is 100 milliseconds,

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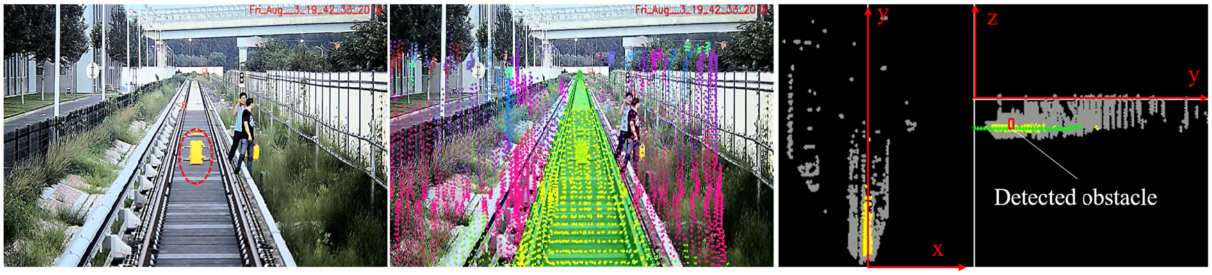
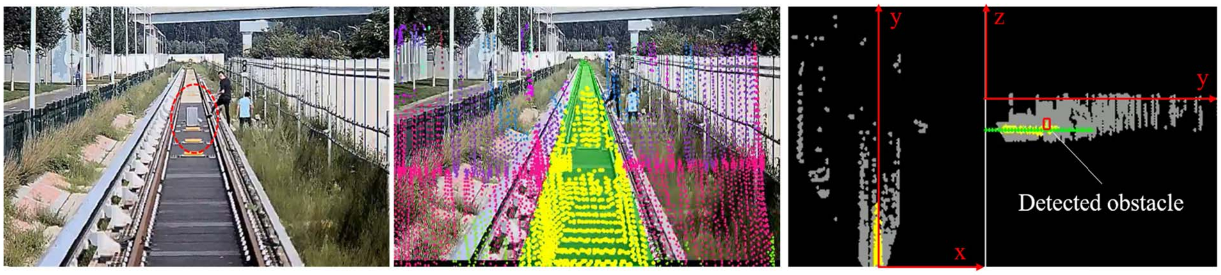
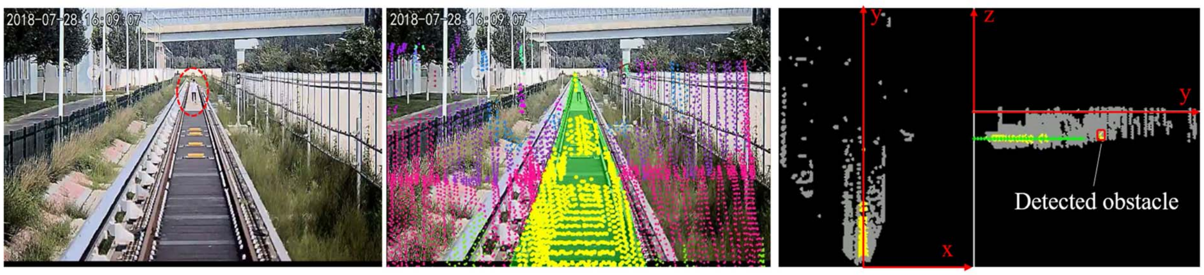


a. Toolkits detection



b. Suitcases detection



c. Pedestrians detection

Fig. 11. Small obstacles detection results. Column 1 are the original images; Column 2 are the fused images with LiDAR data, and yellow points represent the LiDAR points mapped in the railway area; Column 3 are the detection results in the raw LiDAR point cloud from both vertical and side views.

the real-time performance of the proposed network can meet the requirements for practical applications.

C. Objects Detection

1) Small Obstacles Detection: Since it is rare in practice to find small obstacles on the railway track, the obstacles with different sizes were manually set on a straight railway test line for verification purpose.. Toolkits, suitcases and pedestrians were selected to represent the obstacles that the train might encounter when running. The widths and heights of each type of obstacles were 39cm · 58cm, 42cm · 68cm, 40cm · 178cm, respectively.

A series of experiments were conducted to examine whether the proposed method could accurately detect the distance to small obstacles. The obstacles were placed at different test points with known distances (ranging from 12m to 120m) measured by a tape measure. At each test point the experiments for each type of obstacles were conducted ten times. The mean absolute error (MAE) was used as the measure of detection accuracy by our proposed method. The results are shown in Table III.

The symbol "×" in Table III indicates that we could not detect the obstacle at this distance. It shows that the maximum

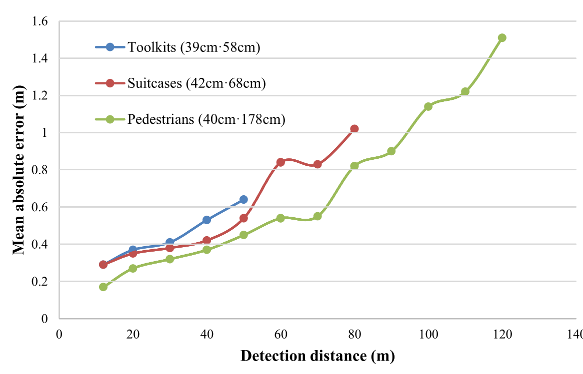


Fig. 12. Detection errors for different obstacles.

detection distance of toolbox, suitcase and pedestrian are 50m, 80m and 120m, respectively. To better visualize the results in Table III, the detection errors for different obstacles were illustrated by referring to the detection distance, as shown in Figure. 12.

It can be found that with an increase in the obstacle size, the detection distance increases; and at a certain distance (e.g., ranging from 12m to 50m), the detection error decreases as the obstacle size increases. It is consistent with the reality since for larger obstacles LiDAR will obtain more reflection

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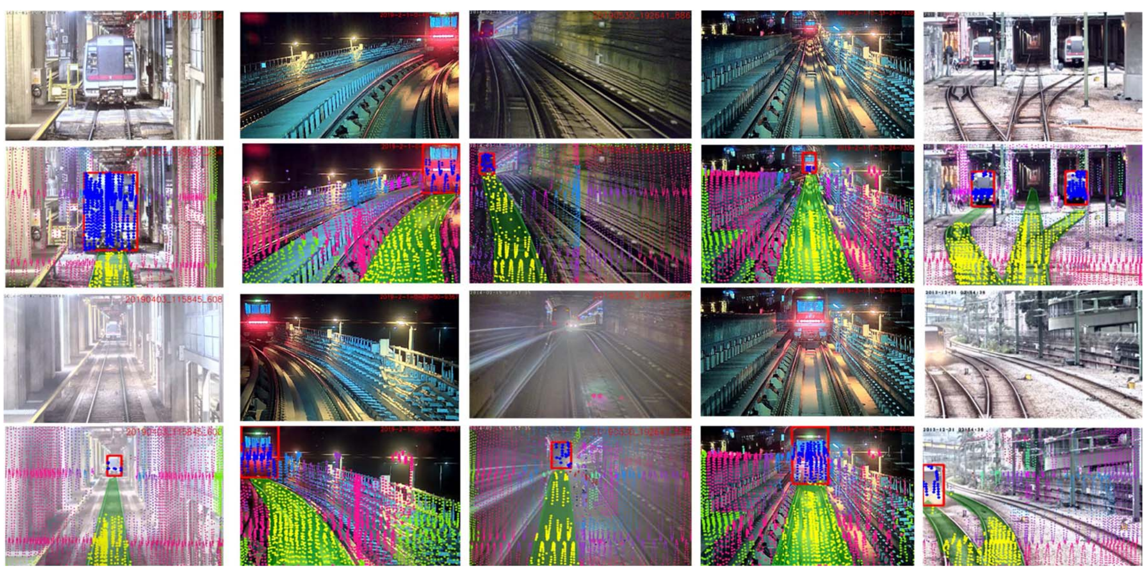
TABLE III
SMALL OBSTACLES DETECTION RESULTS

Real distance (m)	Mean absolute error		
	Toolkits (39cm · 58cm)	Suitcases (42cm · 68cm)	Pedestrians (40cm · 178cm)
12	0.29	0.29	0.17
20	0.37	0.35	0.27
30	0.41	0.38	0.32
40	0.53	0.42	0.37
50	0.64	0.54	0.45
60	×	0.84	0.54
70	×	0.83	0.55
80	×	1.02	0.82
90	×	×	0.90
100	×	×	1.14
110	×	×	1.22
120	×	×	1.51

TABLE IV
FORWARD TRAIN DETECTION RESULTS

	Straight	Curve	Uphill	Downhill	Turnout	Average
TPR	99.3%	99.5%	98%	97.5%	97%	98.4%
FPR	0.67%	1%	1%	2%	2%	1.33%



a. straight b. curve c. uphill d. downhill e. turnout

Fig. 13. Forward train detection results. Rows 1 and 3 are the original images; Rows 2 and 4 are the detection results.

points for detection. On the other hand, for a certain type of obstacle, with the increase of distance the detection error increases gradually. As the surface of railway track is flat, the number of LiDAR points reflected from the rail track will decrease as the increase of distance. Thus, at a long distance there are few LiDAR points on rail tracks and the fitted curve of the railway track plane may incur significant errors. This will lead to inaccurate estimation of the height of the railway track plane, and further the distance to the detected obstacles.

2) Forward Train Detection: We further analyzed the performance of forward train detection. 1000 frames of synchronized LiDAR points and images from the Rail-dataset were selected to evaluate the detection accuracy. The test data covered typical scenes of train operation, such as straight, curve, uphill, downhill and turnout. In our experiment the detection was made for each frame, and it was determined to be a correct case if the percentage error between the detected distance and the real distance to the train is less than 5%. Accordingly, two measures, i.e., true positive rate (TPR) and false positive rate

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TABLE V
COMPARISON OF FORWARD TRAIN DETECTION

Distance (m)	Camera based method		LiDAR based method		**Proposed fusion method**	
	Estimate (m)	Percentage error	Estimate (m)	Percentage error	**Estimate (m)**	**Percentage error**
60	57.6	-4%	58.9	-1.83%	**59.7**	**-0.5%**
80	82.2	2.75%	80.8	1%	**80.6**	**0.75%**
100	103.7	3.7%	×	×	**101.1**	**1.1%**
120	124.2	3.5%	×	×	**119.7**	**-0.25%**
140	147.8	5.5%	×	×	**142.2**	**1.57%**
160	168.2	5.1%	×	×	**164.3**	**2.69%**
180	191.3	6.28%	×	×	**177.4**	**-1.44%**
200	213.6	6.8%	×	×	**202.9**	**1.45%**
220	241.6	9.82%	×	×	**223.1**	**1.4%**
240	258.2	7.59%	×	×	**243.5**	**1.46%**

(FPR), were defined for performance evaluation. Specifically, TPR is the proportion of the cases when trains were detected correctly; and FPR is the proportion of the cases when trains were detected with larger percentage errors than 5% or even if not existing. The detection results are shown in Table IV.

It can be seen that the proposed method achieves high accuracy, i.e., 98.4% and 1.33% in terms of average TPR and FPR, respectively. Compared to the scenarios of straight and curve, the detection accuracy decrease slightly in the scenarios of uphill, downhill and turnout. One possible reason is that when the LiDAR points were mapped to the image in such scenarios, some LiDAR points outside the train might be erroneously included. To better visualize the detection results, some representative illustrations are provided in Figure. 13.

To further demonstrate the advantage of the proposed fusion method, camera and LiDAR were used separately for train detection. A series of experiments were conducted using two trains on a real railway track. The examined distance ranged from 60m to 240m. For the experiments only using the camera, we detected the train using the segmentation algorithm as in Section 3.2, and measured the distance to the forward train using the monocular camera ranging algorithm [42], which is widely used in vehicle Advanced Driver Assistance Systems (ADAS). The results in Table V reveal that the detection only using the camera has undesirable accuracy in terms of percentage error. For the detection only using LiDAR sensor, it was found difficult to classify the LiDAR points belonging to the train. To enable the comparison, we referred to the method in [23] by setting up a cube detection area in front of the train. As the width of the train track is 1.43m, the height of the train is usually more than 4m, and the measurement range of LiDAR is 300m, the height, width and length of the cube detection area is set as 4m, 1.43m and 300m, respectively. In the cube area, the LiDAR points with the same height as the train were classified as the ones belonging to the forward train. The experimental results show that in the scene of straight track the forward train could be detected with high accuracy in a certain range (less than 100m) based on the LiDAR sensor alone. However, in the scenes of curve or downhill, the LiDAR based method often resulted in false detection. Thus, only the reliable detection results (for straight scene within 100m) were presented to show the performance of the LiDAR based method. By comparison, for the proposed fusion method, since both the pixels of the forward train and the corresponding LiDAR points were obtained and utilized, accurate detection could be made even at a longer distance, e.g., 240m.

VI. CONCLUSION

In this study, a railway object detection method was proposed by fusing camera and LiDAR data. The MSP framework enables to detect both large objects such as forward trains and small objects on the railway track such as toolkit, suitcase and pedestrians. The experiments show that the MSP network achieves accurate segmentation of railway track and train with 95.91% MIoU and 97.82% MPA. For small obstacles detection, the toolkit (39cm · 58cm), suitcase (42cm · 68cm) and pedestrian (40cm · 178cm) on the railway track could be detected accurately up to 50m, 80m and 120m away, respectively. For forward train detection, the proposed fusion method achieved the average TPR and FPR of 98.4% and 1.33%, respectively. Besides, it was demonstrated that the fusion method apparently outperformed the one based on camera or LiDAR alone. All these help verify the effectiveness of the proposed method.

However, due to the lack of accurate location information, we could not make full use of prior information of train routes. Future work will be focusing on the enhancement of obstacle detection by integrating the positioning information. Besides, the railway detection in this study relies on semantic segmentation, future work will integrate the results of Hough transform and semantic segmentation to make the railway track detection results more reliable. On the other hand, the extraction of the detection region in this study was highly dependent on camera, making the region of interest extraction affected by light changes. Thus, it is expected to explore the

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method based on LiDAR for railway track and train detection in order to reduce the interference of lighting conditions.

Acknowledgment

The authors would also like to thank the insightful and constructive comments from anonymous reviewers.

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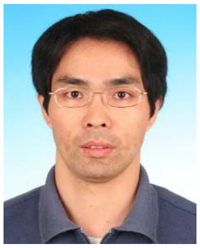

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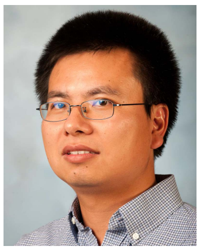

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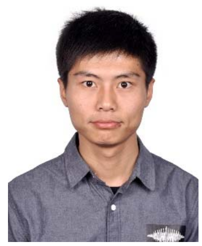

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